

# Science Teaching Reference Handbook

## Physics

Distance is total path length while displacement includes direction from start to finish. Speed is distance per time, while velocity includes direction. Acceleration describes change in velocity over time.

Newton's first law describes persistence of motion when net force is zero. The second law connects net force, mass, and acceleration. The third law describes equal and opposite interaction forces acting on different objects.

Work involves force acting through a displacement. Kinetic energy depends on motion, while gravitational potential energy depends on position in a gravitational field.

Physics lessons should include units and interpretation. A correct numeric answer without units is incomplete in many contexts.

## Biology

Cells contain structures with specialized functions. The membrane controls movement of substances, the nucleus contains genetic material in eukaryotic cells, and mitochondria are involved in cellular respiration.

DNA contains genetic information. During transcription, a DNA template is used to make RNA. During translation, ribosomes read mRNA codons to build a protein.

Photosynthesis stores energy in chemical form using light, carbon dioxide, and water. Cellular respiration releases usable energy from chemical fuels. The processes are related but should not be described as simple opposites in every detail.

## Chemistry

Atomic number is the number of protons. Isotopes of the same element have the same proton count but different neutron counts. Ions form when the electron count changes relative to the proton count.

Chemical equations represent rearrangements of atoms. Balancing changes coefficients, not the chemical formulas of the substances.

Acids and bases can be introduced through pH and observable behavior. Safety matters when discussing real substances, and classroom examples should not encourage unsupervised experiments.

## Science activities

Ask learners to predict, observe, and explain. A short table of measurements can be more useful than a long lecture. Activities should distinguish evidence from explanation.

Diagrams and formulas should be paired with plain-language interpretation so learners understand what the symbols represent.